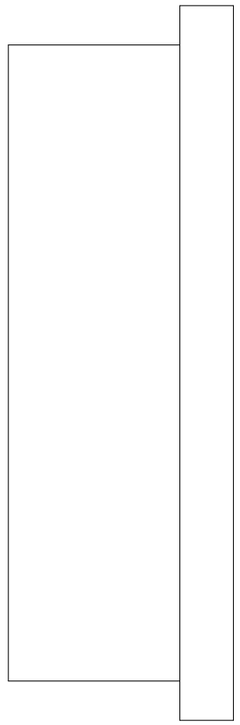
# VAYN shoulder · milling drawing · 33 Bearing cartridge and ring-2

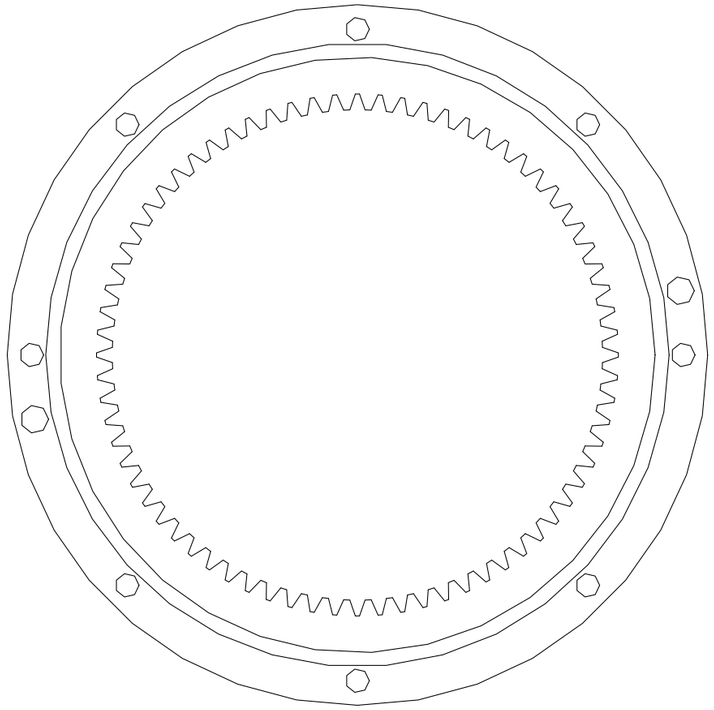


FRONT · Z horizontal, +X up



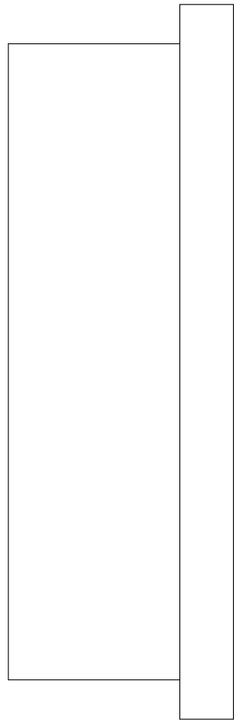
scale 1.17:1 view span 31.5 × 100.0 mm

END · from the arm side (+Z)



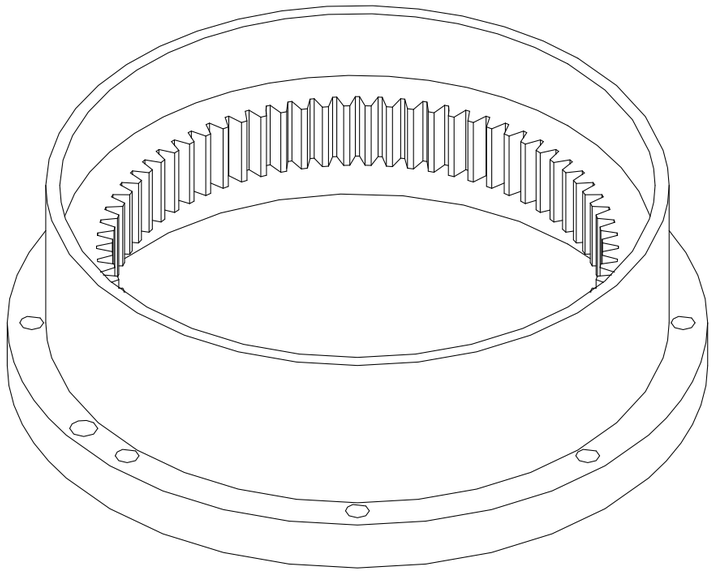
scale 1.14:1 view span 100.0 × 100.0 mm

SIDE · Z horizontal, +Y up



scale 1.17:1 view span 31.5 × 100.0 mm

ISOMETRIC



scale 1.14:1 view span 100.0 × 80.2 mm

## 33

## Bearing cartridge and ring-2

Quantity in this actuator: 1

Material: 20MnCr5

Group: Gears

Finished size: Flange Ø100 × 7.5 mm, spigot Ø89 mm, overall 31.5 mm

Mass: 350.87 g each

Model bounding box, X × Y × Z: 100.0 × 100.0 × 31.5 mm

Raw stock: Same 20MnCr5 bar

Process: Turn, wire-EDM the ring-2 teeth while the steel is still soft, drill and tap

Heat treatment / finish: Carburize the ring-2 teeth only. Case 0.15–0.20 mm, 58–62 HRC. Tap the 8 × M2.5 first and mask them. Mask the Ø89 spigot and the Ø85 bore. No tooth grind. Finish Ø89 and Ø85 after heat treatment.

STEP solid: 33 Bearing cartridge and ring-2 (20MnCr5)

**Controlled features**

- Spigot Ø89 h6.
- Bearing bore Ø85 is selective: 3–10 µm larger than the measured 61813 outer diameter. The rear ring must still slide.
- Ring-2: 72 teeth, module 1.0. Between Ø1.5 mm pins, Mi = 70.499 mm, plus 0.015 / minus 0.
- 8 × M3 counterbored holes on Ø93 mm. 2 × Ø4 mm dowel holes.
- 8 × M2.5 tapped for the cover, before carburizing, and masked.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 33_cartridge.pdf